

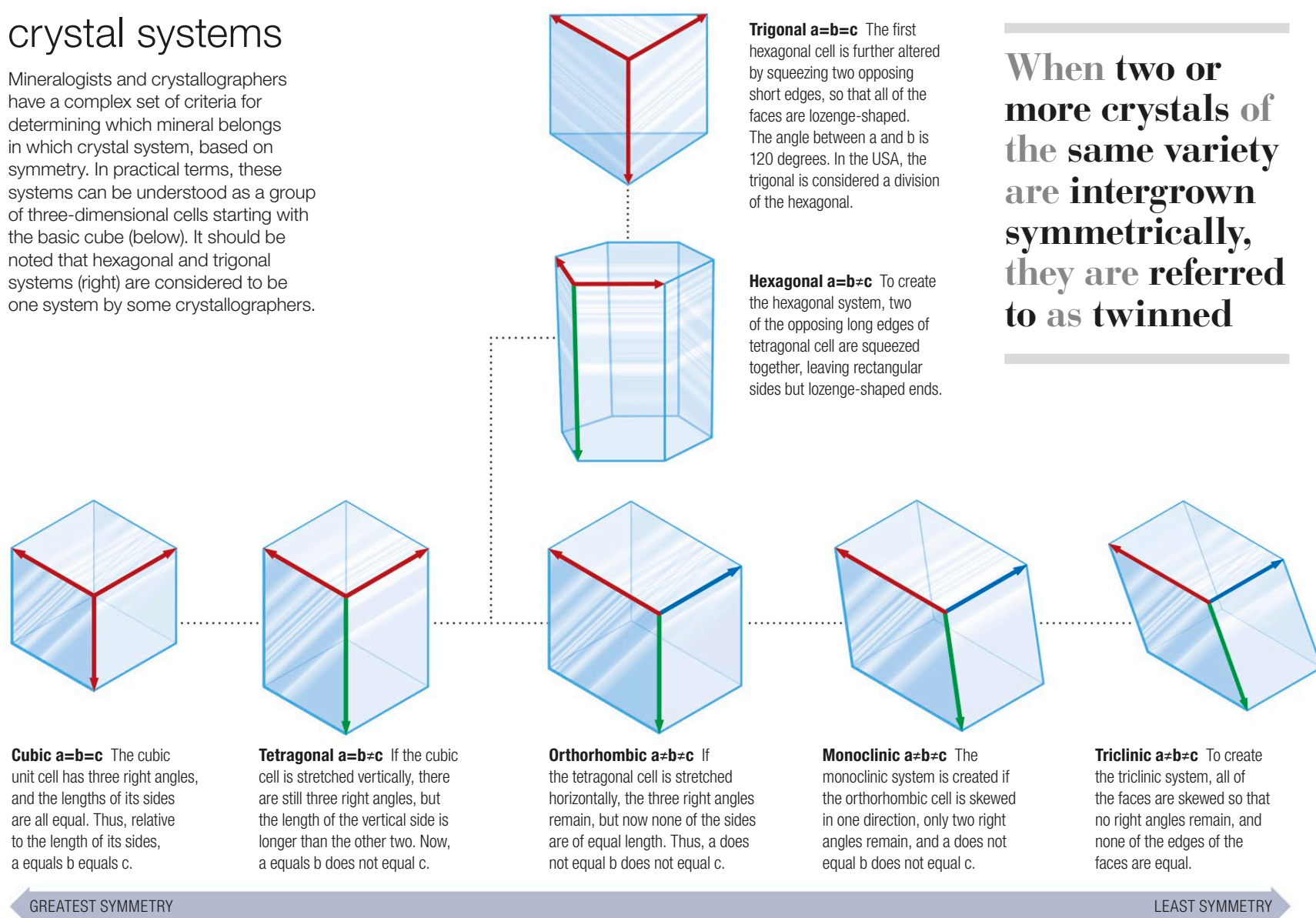
Crystal systems and habits

A crystal is a solid in which the component atoms are arranged in a particular, repeating, three-dimensional pattern. When these internal patterns produce a series of external, flat faces arranged in geometric forms, this forms a crystal. These repeating structures are identical structural units of atoms or molecules, and are called unit cells. The unit cell is reproduced over and over in three dimensions, meaning that the shape of the crystal will resemble that of the individual unit cell. The

crystals of different minerals can have unit cells that are the same shape but are made of different chemical elements. Because a crystal is built up of repeating geometric patterns, all crystals exhibit symmetry, depending on the basic geometry of their unit cells. These fall into seven main groups and are called crystal systems. The final external form a crystal takes is known as its habit, and the shape produced by a mass of numerous identical crystals is a growth habit (see opposite).

Minerals and crystal systems

Mineralogists and crystallographers have a complex set of criteria for determining which mineral belongs in which crystal system, based on symmetry. In practical terms, these systems can be understood as a group of three-dimensional cells starting with the basic cube (below). It should be noted that hexagonal and trigonal systems (right) are considered to be one system by some crystallographers.



When two or more crystals of the same variety are intergrown symmetrically, they are referred to as twinned